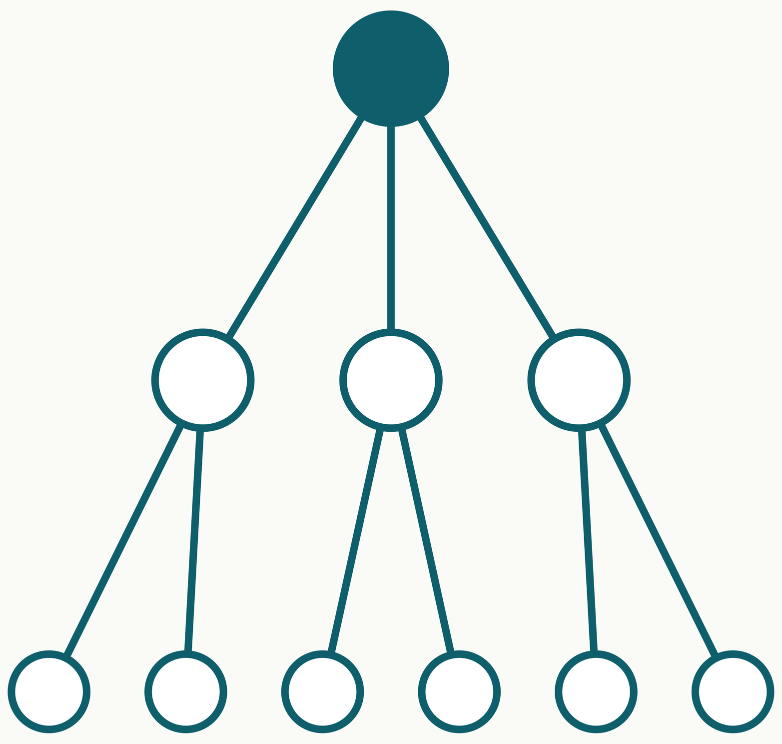
The Bruhat–Tits Tree Code — Status and Test Plan

Rowan Brad Quni-Gudzinas · QNFO · CWI Summer School on Quantum Algorithms & QEC · Wed 26 Aug 2026, 16:30–18:00

Why it matters

Fault tolerance today means surface codes: many hundreds of physical qubits per logical qubit, active syndrome measurement, millikelvin cooling. The tree code is a different geometry whose **simulated** thresholds run 4.6–75× above surface-code figures — and whether that survives a matched-noise comparison is test T1. If it does, the payoff is energy: fewer joules per correct answer.



The Bruhat–Tits tree T_2 — the geometry behind the code. Every internal vertex carries a [[3,1,1]] perfect tensor.

Verified

Thresholds computed three ways — analytical recursion, exact stabilizer simulation, Monte Carlo — and re-checked by a deposited 35-check script.

10.5281/zenodo.20109836 · 10.5281/zenodo.22038733

Boundary

Under independent errors the qudit family sits at ≈ 2×10^{-4}, about 55× below surface codes. The advantage claim targets correlated-failure regimes — exactly what T1 probes.

10.5281/zenodo.21046993 · 10.5281/zenodo.22025544

Not yet

No hardware run. Tests T3–T4 are the instrumentation path.

The code and the numbers

A holographic perfect-tensor code on a truncated Bruhat–Tits tree: a [[3,1,1]] perfect tensor sits at every internal vertex; logical information lives at the root and physical qubits at the leaves. Each block is a one-to-two-qubit isometry with distance d = 1 — protection emerges from concatenation across the tree, not from any single block. Error correction is built into the geometry rather than applied by active measurement loops.

Error model (qubit, code-capacity)	Tree code	Surface code	Ratio
Bit-flip	**50.0%**	~10.9%	4.6×
Depolarizing	**75.0%**	~1.0%	75×
Independent X+Z	**17.30%**	—	—

Classical simulation only: i.i.d. noise, crossing criterion; surface comparators are the standard published figures (bit-flip ~10.9% code-capacity, depolarizing ~1% circuit-level). No tree code has run on quantum hardware yet.

The tests that would change my mind

T1 · Matched-noise comparison

Tree vs surface under the *same* correlated noise model. Classical, days–weeks.

Would change my mind: the advantage falls to parity.

T2 · Decoder + full overhead

Physical qubits plus classical decoding cost. Classical.

Would change my mind: the advantage disappears once decoding cost is counted.

T3 · Random walk on a p-regular tree

Return-probability decay. Ion / analog simulator, ~8 weeks.

Would change my mind: no change in the decay exponent where predicted.

T4 · Clock-rest split

Diagonal 0% vs non-diagonal 29–35% violation rate. Trapped ion, ~8 weeks.

Would change my mind: the measured split falls outside that range.

T5 · Independent re-run

An outside group repeats T1 with my simulator and the protocol.

Would change my mind: nothing to predict — this one is about reproducibility.

1. A correlated-noise benchmark — model and data — I can run T1 against.
2. A decoder reality check: does a known decoder family already cover tree-structured codes, and how does it scale?
3. Pointers on perfect tensors for p > 2 (the binary [[3,1,1]] case is settled).
4. An independent re-run of T1.

The five tests are the takeaway: every criterion is stated with numbers, so a null result is a result. The simulator and the protocol are handed over to anyone who wants to run T1 or T5 themselves.

All numbers are classical simulation — i.i.d. noise, crossing criterion; nothing has run on hardware yet. · DOIs (verified 21 Aug 2026): 10.5281/zenodo.20109836 (architecture) · 10.5281/zenodo.22038733 (guide + verification script) · 10.5281/zenodo.21046993 (qudit QEC) · 10.5281/zenodo.22025544 (trapped-ion testbed) · 10.5281/zenodo.21964674 (QEC–Darwinism audit) · 10.5281/zenodo.21979060 (prime-valuation correction) · Full plan: github.com/QNFO/cwi-qec-poster-2026 → docs/tree-code-roadmap.md